

Reducing the dose and rate of infusion reduces inflammatory reactions in patients given IV erythromycin [5], and may be extrapolated to other macrolide infusions.

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Discounted cash flow of anesthesia information management systems

To the Editor:

Anesthesia Information Management Systems (AIMS) increase the reimbursement for operating room (OR) services by reducing omissions, facilitating coding and transfer of an electronic record, prompting physician input of missing data, increasing accuracy of data entry, and improving the revenue cycle [1]. Additional benefits include improved staff scheduling, medical decision-making support, and quality improvement monitoring [2]. Advances in the sophistication of computer workstations and network components, and an industry trend towards electronic medical record (EMR) implementation, have driven the development of AIMS that interfaces with the hospital's software system. Despite these advances, fewer than 90% of medical centers have adopted an AIMS due to the uncertainty of various costs [2].

The business decision to purchase an AIMS takes into account the net present value (NPV) of daily cash flows, the total cost of ownership (TCO), and return on investment (ROI) calculations. Total cost of ownership and ROI are limited by their exclusion of the time value of money and a bias against long-term investments, respectively. Total cost of ownership focuses on training, maintenance, and technical support costs, which are uncertain and difficult to estimate. Capital budgeting decisions should be based on the discounted cash flow method [3]. Net present value estimates the value of the AIMS's projected future cash flow as if the cash flows were available today. The NPV of daily operating activities is calculated using the change in net working capital, operating cash flow (OCF), and capital spending [4]. The elements of change in net working capital are inventory, accounts receivable, and accounts payable. Operating cash flow is estimated from the cost savings in anesthetic-related drug costs, monetary savings per case, and hospital reimbursement increases. Capital spending includes the cost of the AIMS software, servers, workstations, routers, and cables. Table 1 lists the determinants of OCF and capital spending.

Consider the Memorial Healthcare System, which performs 9,200 procedures in 14 ORs and generates \$71,901,048 in revenue. Senior managers are considering adopting an AIMS at 17 workstations within the OR, postanesthesia care unit, and business office. An approximation of the net present value to the Memorial Healthcare System is determined by considering published estimated figures. Operating rooms account for approximately 40% of a hospital's revenue [5]. A \$32 per case reduction in anesthetic drug costs, a \$100 per case cost savings, and a 1.5% increase in hospital reimbursement were observed following installation of an intraoperative data management system [6-8]. Cost estimates for the AIMS software, servers, workstations, routers, and cables are proposed [9]. Assuming that net working capital remains unchanged by the purchase, an AIMS generates a cost savings to the Memorial Healthcare System of \$1,643,860 per year and a capital cost of \$312,000 (Table 2). The NPV of the AIMS is positive, indicating that the purchase would add \$523,655 in value to the Memorial Healthcare System (Table 3).

Table 1 Determinants of cash flow and capital spending

1. Operating cash flow
 - i. anesthetic-related drug costs:
 - number of cases per year
 - ii. money per case:
 - number of cases per year
 - iii. hospital reimbursement:
 - gross revenue
2. Capital spending
 - i. number of operating room work stations
 - ii. number of office work stations
 - iii. estimate of routers/switches/cables

Table 2 Operating cash flow and capital investment

3. Operating cash flow

i. anesthetic-related drug costs:

- reduced anesthetic-related drug costs = 9,200 cases per year × \$32 per case = \$294,400

ii. money per case:

- financial saving per case = \$100 per case × 9,200 cases per year = \$920,000 per year

iii. hospital reimbursement:

- annual increase: $\$71,910,048 \times 0.4 \times 0.015 = \$431,460$ per year

Total cost savings per year = \$294,400 + \$920,000 + \$431,460 = \$1,643,860 per year

4. Capital spending

i. number of operating room work stations

- \$8,000 per operating room work stations × 14 operating rooms = \$112,000
- \$10,000 per AIMS and server software × 14 workstations = \$140,000

ii. number of office work stations

- \$5,000 per recovery room and office workstation × 3 workstations = \$15,000
- \$10,000 per AIMS and server software × 3 workstations = \$30,000

iii. estimate of routers/switches/cables

- routers, switches, and cables = \$15,000

Capital Investment: \$112,000 + \$140,000 + \$15,000 + \$30,000 + \$15,000 = \$312,000

AIMS = anesthesia information management systems.

Table 3 Net present value (NPV) and return on investment

$$NPV = [OCF / (1+R)^t] - C$$

where OCF is operating cash flow, R is cost of capital = 7%, t is time = 10 years, and C is capital investment

$$NPV = [\$1,643,860 / (1+0.07)^{10}] - \$312,000$$

$$NPV = \$835,655 - \$312,000$$

$$NPV = \$523,655$$

From a financial perspective, a positive NPV indicates that an AIMS adds value to a health system and should be purchased. However, the purchase decision should not be based exclusively on the NPV. Many nonfinancial factors support the implementation of an AIMS such as improvements in the quality of perioperative care, improvements in the management of intraoperative data, and increases in hospital revenue [10]. Clinicians and administrators may promptly analyze the financial benefits of an information system. Knowledge of specific determinants of cash flow and capital costs facilitate the discounted cash flow analysis of AIMS.

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A novel way of anesthetizing and maintaining airway/ventilation in an ultra-morbidly obese patient presenting for upper GI endoscopy[☆]

To the Editor:

Morbid obesity is encountered increasingly in ambulatory surgery centers (ASC), including patients presenting for gastrointestinal (GI) endoscopy. At one university teaching hospital, about 6% to 7% [1] of all patients presenting for endoscopy had a body mass index (BMI) of more than 40 kg/m². Frequently these patients present to the endoscopy center for a gastroscopy prior to weight loss surgery. We describe the anesthetic management of one such patient with a BMI of 79.5 kg/m².

A 41 year old, 281 kg, 188 cm man presented for an upper GI endoscopy. Significant past history included well controlled asthma and shortness of breath with chest pain on moderate exertion. He was also recently diagnosed with sleep apnea and was using continuous positive airway pressure. Past surgical history included successful inguinal hernia repair and tonsillectomy. Current medications included ipratropium (Atrovent) and albuterol for asthma. Among significant physical examination findings were a loose molar and few stable false teeth. He had a Mallampati class 3 airway. His neck extension and flexion were both good.

Normally these patients undergo endoscopy in the left lateral position. However, we decided to perform the procedure in supine head-up position. Anesthesia equipment included a modified Mapleson C breathing system, nasal airways

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